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Note: This paper is a draft for Chapter 5 of my LL.D. dissertation. To view the latest version of this paper in its appropriate context, please visit https://github.com/emersonbanez/dissertation_public

1 Overview of Encoding and Analysis Approaches

2 - Normative Component

3 In order to carry out automated reasoning of law, we have to encode legal norms into
4 computational forms. In the previous section, ontologies and description logic, helped
5 us define the descriptive component of the legal knowledge (in this case the OECD
6 Competitive Impact Assessment tests) that we seek to encode. The analysis that can
7 be performed on an ontology-based data structure can reveal implicit relationships
8 between entities (such as inheritance, equivalence), as well as inconsistencies. However,
9 ontologies only give us a part of the picture. Besides the entities, their attributes and
10 interactions - all these are subject to constraints and transformations based on law.
11 These only provide static data about the semantics of entities and their interactions -
12 but these do not reflect the legal constraints that act upon those objects, and how the
13 semantics could be qualified, transformed, or annulled by such constraints. Another
14 way of putting it is that ontologies reflect only the whats and the whos, not the oughts
15 and ought nots that are contained in legal knowledge.

16 In this chapter, we shall cover the requirements of a logical system for representing
17 important normative features of a body of rules: First, that it should capture the
18 conditional nature of legal inferences; Second, it should involve modalities other than
19 True or False - that is, it should work on normative states (for example, whether
20 propositions are permitted, forbidden, or obligatory); Finally, it should also allow for
21 the possibility of inferences being defeated by additional information. The chapter
22 shall describe these features in turn, and propose Reified IO Logic as an encoding
23 system that integrates all these requirements.

24 The choice of encoding system is based on Robaldo (2020)’s description of a compu-
25 tational knowledge base for legal rules,¹ which accomodates several levels of encoding:

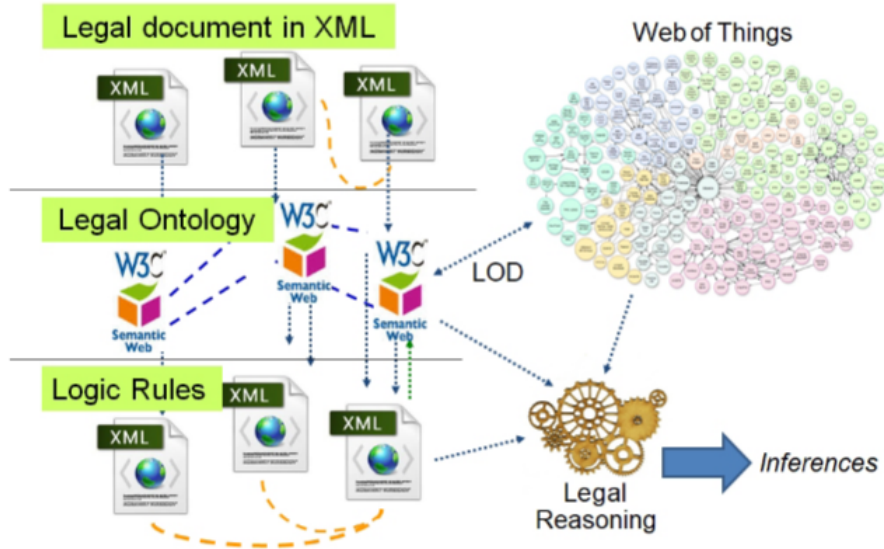
26 1. **Legal text** - Written in a human readable language but tagged and structured
27 through an XML-based markup (such as LegalDocML, an OASIS standard for
28 legal markup). At this level, the system designer encodes the law as is, but
29 provides markup for some sections in order to signal the structure of the docu-
30 ment, as well as highlight concepts that are relevant to the ontology and logic
31 layers. This allows systems to associate these elements with subsequent logical
32 encodings the represent their meaning. This will allow components of the text
33 to be linked to the subsequent encodings and support automated processing.

34 2. **Legal ontology** - This consists of the formalized naming and definitions of
35 concepts that are contained in the human-readable rules, as described in the
36 previous chapter. Concepts and relationships are encoded in OWL, and will
37 serve as the predicates to be used in Thehe normative logic layer. The ontology
38 can also be described in terms of Description Logic, and can support some
39 analysis. Howeverm this ontology layer alone is not fit for legal reasoning,
40 as it does not account for deontic aspects of the rules, or accounts for their
41 defeasibility.

42 3. **Normative logic** - This layer represents the normative content of the rules,
43 represented as logical formulae. This logic layer is formalized in a defeasible
44 form of deontic logic and then encoded in LegalRuleML.

¹Livio Robaldo, Cesare Bartolini, and Gabriele Lenzini. “The DAPRECO Knowledge Base: Representing the GDPR in LegalRuleML”. in: *Proceedings of the 12th Conference on Language Resources and Evaluation (LREC 2020)*. Marseille, May 11–16, 2020, pp. 5688–5697, at 5688-5689.

Figure 1: Layered Architecture of Encoding(Robaldo, 2020)



1.1 Availability of Multiple Logical Systems

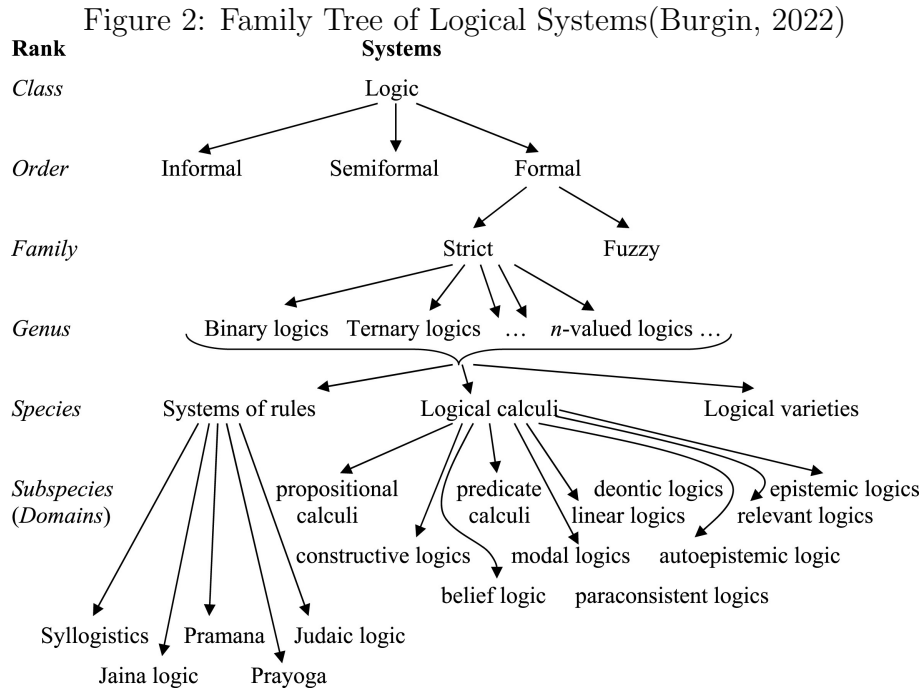
In stating that we will translate legal rules into a logical encoding, we mean “logic” as a formal method that can support deductive reasoning. That is, proving a conclusion by means of at least two other propositions.² The term includes not just Aristotelian syllogism, but can accommodate other forms of deductive inferences, such as the logic of alternatives, compound propositions, and of relationships, and the study of propositions themselves.³ Burgin (2022) provides a high-level overview of the evolution of logical systems: From loose collection of rules related to belief systems to more modern, formalized logics.⁴ The diversity of logics can provide tools

²Ruggero Aldisert, Stephen Clowney, and Jeremy Peterson. “Logic for Law Students: How to Think Like a Lawyer”. In: *University of Pittsburgh Law Review* 69 (2007), pp. 1–22, at 2.

³Anthony G. Guest. “Logic in the Law”. In: *Oxford Essays in Jurisprudence*. Oxford: Oxford University Press, 1961, pp. 176–197.

⁴See generally Mark Burgin. “Evolution of logic as an information processing mechanism in advanced biological systems”. In: *Bio Systems* 221 (2022), p. 104758. ISSN: 0303-2647. DOI: 10.1016/j.biosystems.2022.104758. URL: <https://www.sciencedirect.com/science/article/pii/S0303264722001393>.

for the representation of various aspects of knowledge. Each logic can capture and emphasize a certain level of description, or comprehend specific problems.



The development of novel logical systems now allow us to have a more focused view of a problem.⁵ For example: Triadic logics which allows for intermediate truth values, rejecting the law of the excluded middle; A fuzzy logic, which has instead of True or False, an infinite continuum of possible values, along with a more informal process of inference. Then there are deontic logics, with new operations such as “obligatory”, “permitted”, and “forbidden”.⁶

⁵Susan Haack. “On Logic in the Law: “Something, but Not All””. In: *Ratio Juris* 20.1 (2007), pp. 1–31. ISSN: 0952-1917, 1467-9337. DOI: 10.1111/j.1467-9337.2007.00330.x. URL: <https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9337.2007.00330.x> (visited on 03/26/2024), at.

⁶Deontic logics in particular have attracted legal theorists as a way to make formal, rigorous representations of the structure of legal orders. A variation of the idea of deontic modes is a component of the encoding sytem proposed later in this chapter *ibid.*, at 11-12.

1.2 Legal Norms as Conditional Inferences

In more conventional forms of logic, we can readily represent factual statements. Take for example the proposition, earlier made explicit as a fact in the ontology, that corporations are also persons:

$$\text{All Corporations are Persons}(\text{All } S \text{ is } P) \quad (1)$$

The predicate Persons is descriptive of the subject Corporations. If the proposition is admitted, it follows that no Corporations are not Persons. If the proposition is denied, then it follows that some Corporations are not Persons.

However, let us take a normative or legal statement, “Any person who shall abuse its market dominance shall be guilty of a criminal offense”. The predicate “guilty of a criminal offense” is not necessarily descriptive of the subject “person who abuses market dominance”. The relationship between the components of the proposition hinges on the injunctive “shall”, which is not descriptive of what is, but instead denotes what ought to be under certain contingencies.⁷ Normative propositions are more comparable to the structure of causal inferences: If p then q . Instead of making factual predictions, however, they are statements of what ought to be.⁸ Sartor states that legal provisions may be restated into such atomic inference rules, with the basic if p then q structure.⁹ It may also be possible to state a rule categorically as simply

⁷Guest, “Logic in the Law”, at 183-184.

⁸Guest also clarifies that these are not necessarily imperative statements or commands. *ibid.*, at 184.

⁹The consequent of each rule is a literal; and the antecedent is a conjunction of literals. A literal is an atomic formula or the negation thereof. A positive literal has the form ‘ $p(x)$ ’ where ‘ p ’ is a predicate symbol and ‘ x ’ is a list of terms. On the other hand, a negative literal has the form ‘not $p(x)$ ’ where ‘not’ is a logical negation. The complement $\bar{q}$ of a literal q denotes the literal opposed to q .

80 q , but this should be rare in operation.¹⁰.

81 Take for example rule A1 in the OECD Guidelines, that a provision should be
82 flagged if it “Grants exclusive rights for a supplier to provide goods and services”.
83 This can be broken down to several inference rules:

If [Law Requires Single Supplier] then [Rights are Exclusive] (2)

If [Rights are Exclusive] then [Flagged] (3)

84
85 Where [Rights are Exclusive] stands for p and [Flagged] stands for q .

86 Initially, we can think of rules of substantive law as statements of the specific factual
87 conditions upon which specific consequences depend. The applicability of the condi-
88 tion can be contingent on several conditions, as well as the absence of exceptions.¹¹
89 Thus:

: If q is a positive literal p , then $\bar{q}$ represents the negative literal not p ; if q is the negative literal not p , then $\bar{q}$ represents the positive literal p . Giovanni Sartor. “A Formal Model of Legal Argumentation”. In: *Ratio Juris* 7.2 (July 1994), pp. 177–211. ISSN: 0952-1917, 1467-9337. DOI: 10.1111/j.1467-9337.1994.tb00175.x. URL: <https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9337.1994.tb00175.x> (visited on 06/20/2023), at 179.

¹⁰Sartor’s formalization also admits of these so-called “degenerate inference rules”. These enable unconditional derivation of any instance of their conclusion A. These categorical inference rules can be used to express some forms of ungrounded assertion, such as: 1. Statements of undisputed empirical evidence (the facts that justify a law or court decision); 2. Basic (and very general) normative postulates; 3. Tentatively advanced propositions for which no ground is currently available *ibid.*, at 179.

¹¹Jerome Michael and Mortimer J. Adler. “The Trial of an Issue of Fact: I”. in: *Columbia Law Review* 34.7 (Nov. 1934), pp. 1224–1306. ISSN: 00101958. DOI: 10.2307/1116103. JSTOR: 1116103. URL: <https://www.jstor.org/stable/1116103?origin=crossref> (visited on 12/11/2024), at 1241.

If events $x_1 \dots x_n$ is the case, and unless there are $y_1 \dots y_n$, then z is the case.

(4)

90

91 A legal system can be represented as a body of such propositions that can be eval-
92 uated not based on truth or falsity, but by some other normative standard (such as
93 social benefit, or compliance with other higher rules). Once we have formal represen-
94 tations, the next step would be to apply analytical methods grounded in logic. We
95 can trace the chain of inferences (e.g. via *modus ponens*), discover other rules, even
96 look for potential inconsistencies.¹²

97 1.3 Deontic Logic

98 Legal statements are for the most part, not composed of factual statements. They
99 do not describe the state of the world as it is, but how it ought to be. They cannot
100 be assessed for truth values. Logicians have since developed a form of logic, called
101 Deontic Logic, which is not concerned with True or False, but oughtness. Although
102 notions of Deontic Logic have been explored in fourteenth century Europe as well
103 as Islamic thought (in the 10th century), its modern version grounded in symbolic
104 logic is based on the work of von Wright (1951).¹³ Under von Wright(1951)’s classic

¹²Inference rules are mono-directional, to be used/understood only forward (*modo ponente*) and not backward (*modo tollente*). The consequent q can be derived whenever the antecedent p is satisfied. However, the negation of p cannot be derived when q is assumed to be false. The “if” connective in inference rules is not the same “if” in logical conditionals. Sartor, “A Formal Model of Legal Argumentation”, at 179.

¹³Paul McNamara and Frederik Van De Putte. “Deontic Logic”. In: *The Stanford Encyclopedia of Philosophy*. Ed. by Edward N. Zalta. Spring 2022. Metaphysics Research Lab, Stanford University, 2022. URL: <https://plato.stanford.edu/archives/spr2022/entries/logic-deontic/> (visited on 06/08/2022), at § 1.

105 formulation, it is concerned with the following modes of obligation:¹⁴

- 106 • **Obligatory** - That which we ought to do
- 107 • **Permitted** - That which we are allowed to do
- 108 • **Forbidden** - That which we must not do

109 For von Wright the starting point of his deontic system is the concept of “Per-
110 mitted” as the basic operator - e.g. a proposition can ϕ is Permitted, $P\phi$. Other
111 operators can then be defined in terms of P :¹⁵

$$F\phi =_{df} \neg P\phi - \text{something Forbidden is not Permitted} \quad (5)$$

$$O\phi =_{df} \neg P\neg\phi - \text{something Obligatory is something not Permitted not to do} \quad (6)$$

112
113 Deontic logic was later axiomatized and developed to what is now known as Stan-
114 dard Deontic Logic (SDL). Under SDL, the primary operator is Obligation, denoted
115 as by the symbol $\bigcirc$ (or “ought”). The Permitted operator can be defined as:¹⁶

$$PE\phi =_{df} \neg \bigcirc \neg\phi \quad (7)$$

¹⁴G.H. von Wright. “Deontic Logic”. In: *Mind* 60.237 (Jan. 1951), pp. 1–15, at 1.

¹⁵Donald Nute, ed. *Defeasible Deontic Logic*. Dordrecht: Springer Netherlands, 1997. ISBN: 978-90-481-4874-5 978-94-015-8851-5. DOI: 10.1007/978-94-015-8851-5. URL: <http://link.springer.com/10.1007/978-94-015-8851-5> (visited on 11/23/2024), at 2.

¹⁶Ibid., at 2.

117 That is, ϕ is Permitted if and only if it is not an Obligation that not ϕ . We can
 118 thus construct all the other operators in terms of $\bigcirc$ (See Table 1 at 10 below).¹⁷

Table 1: SDL Definitions of Deontic Operators

Definition	Implication	Example
$\bigcirc(\text{OB})$	A proposition is obligatory if it must occur	It is OBLigatory to pay taxes
$\text{PE } \phi =_{df} \neg \bigcirc \neg \phi$	A proposition is permissible iff (if and only if) its negation is not obligatory	It is PERmitted to drive a car
$\text{IM } \phi =_{df} \bigcirc \neg \phi$	A proposition is impermissible iff (if and only if) its negation is obligatory	It is IMpermissible to smoke in a restaurant
$\text{OM } \phi =_{df} \neg \bigcirc \phi$	A proposition is omissible iff it is not obligatory (can be omitted or not done without violating a norm)	It is OMissible to attend that party (you can attend or not attend)
$\text{OP } \phi =_{df} (\neg \bigcirc \phi \wedge \neg \bigcirc \neg \phi)$	A proposition is optional iff neither it nor its negation is obligatory	It is OPTional to work from home
$\text{NO } \phi =_{df} (\bigcirc \phi \vee \bigcirc \neg \phi)$	A proposition is non-optional iff it is either obligatory or impermissible	It is NON-optional to wear a seatbelt while driving

119 Recasting the earlier example under the Deontic mode:

If [Law Requires Single Supplier] then [Rights are Exclusive] (8)

If [Rights are Exclusive] then $\bigcirc$ [Flagged] (9)

¹⁷McNamara and Van De Putte, “Deontic Logic”, at § 1.2.

121 The inference that the law should be flagged is neither True nor False, but has
 122 the deontic modality of Obligation. Once represented formally, it may be possible to
 123 evaluate a specific statement based on the axioms and theorems of the chosen system
 124 of deontic logic. For example, in von Wright’s classical system, there exists the
 125 Principle of Deontic Distribution which provides that: “If an act is the disjunction
 126 (”or”) of two other acts, then the proposition that the disjunction is permitted is
 127 equivalent to the disjunction of the propositions that the first act is permitted and
 128 the proposition that the second act is permitted”:¹⁸

$$P(\phi \vee \psi) \leftrightarrow P\phi \vee P\psi \quad (10)$$

130 Applying this to the proposition that a Supplier is permitted to supply goods or
 131 services, then the permission is distributed individually to the supply of goods as
 132 well as the supply of services. While this distributive property is a feature of this
 133 particular system of deontic logic, it is not universally accepted, and we can discard
 134 this axiom if it conflicts with our normative intuition.

135 1.4 Defeasible Deontic Logic

136 Another attribute of legal propositions is that they are **defeasible**. This means
 137 that they are tentative - accepted until some other proposition - a new fact that
 138 activates an exception, better evidence, or even a higher law - defeats our original
 139 proposition.¹⁹

¹⁸Nute, *Defeasible Deontic Logic*, at 2.

¹⁹See generally Giovanni Sartor. “Defeasibility in Legal Reasoning”. In: *Rechtstheorie* 24.3 (1993), pp. 281–316.

140 Substantive legal provisions often have a **positive condition**, the event or circum-
 141 stance that must obtain for the purported legal consequence to be arrived at. At the
 142 same time, these conditions are most likely subject to exceptions - elements that ac-
 143 cording to some antecedent norms has to be absent in order for the legal consequence
 144 to apply: The sum of positive conditions embody the determination of the legisla-
 145 tor of what circumstances should normally give rise to the legal consequences. On
 146 the other hand, the exceptions represent special circumstances that can override the
 147 positive conditions, making the legal consequences not applicable. Legal conclusions
 148 are often subordinated structures: The presence of other legal provisions (that are
 149 of equal or higher priority in a hierarchy of norms), which may provide (or negate)
 150 conditions and exceptions.²⁰ The goal of legal reasoning in actual cases is to show
 151 that certain acts, claims, decisions comply or does not comply with the law. This
 152 requires demonstrating that the presence (or absence) of conditions and exceptions.²¹
 153 Thus, legal conclusions are arrived at based on knowledge that is incomplete, uncer-
 154 tain, and inconsistent²² - on plausibility rather than truth. Despite this, an adequate
 155 formalization of defeasible reasoning should provide a sound basis of what to believe.

156 Various such formalizations have been developed to embody defeasibility of reason-
 157 ing. For our purposes, a system of defeasible reasoning should allow for the represen-
 158 tation of various propositions and their attributes: 1. Atomic "facts" that are taken
 159 as a given; 2. Rules (whether or not they are subject to exceptions); 3. Defeating

²⁰This arises from what Stuart Hampshire calls the "inexhaustibility of description" Any situation can embody an inexhaustible set of features, but we can only confront and understand part of it at any given time. See Juan Carlos Bayon. "Why Is Legal Reasoning Defeasible?" In: *Diritto & Questioni Pubbliche* 2 (2002), pp. 1-18, at 3; Citing Stuart Hampshire, ed. *Public and Private Morality*. Cambridge: Cambridge University Press, 1991. 143 pp. ISBN: 978-0-521-22084-2 978-0-521-29352-5, at 30.

²¹Bayon, "Why Is Legal Reasoning Defeasible?", at 3.

²²See Kathleen Freeman and Arthur M. Farley. "A Model of Argumentation and Its Application to Legal Reasoning". In: *Artificial Intelligence and Law* 4.3-4 (1996), pp. 163-197, at 165.

propositions and/or superiority relationships; 4. In the case of legal statements especially, their deontic modal values (Obligatory, Permitted, or Forbidden). A system of defeasible reasoning should also enable operations on these propositions, such as resolving conflicts and making plausible inferences. For example - through prioritization of certain rules and/or the evaluation of supporting or undercutting evidence.²³

In Defeasible Deontic Logic (DDL), legal norms are the positive conditions that prescribe behavior through Permission, Obligation, and Prohibitions. These norms may be subject to exceptions (which are also expressed as norms).²⁴ DDL allows for the representation of facts, defined as whatever can be considered as conclusive unambiguous statements. Facts can include: either a state of affairs or actions already performed (both considered to always hold true). Based on our ontological definitions, we can state that "Acme Inc. is a corporation" through:

$$\text{Corporation}(\text{Acme Inc.}) \quad (11)$$

A rule in DDL is a relationship between a set of antecedents or premises (clauses), represented as $X_1, \dots, X_n$ and the consequent conclusion or conclusion (effect) of the

²³Hanif Bhuiyan et al. "Traffic Rules Encoding Using Defeasible Deontic Logic". In: *Frontiers in Artificial Intelligence and Applications*. Ed. by Serena Villata, Jakub Harašta, and Petr Křemen. IOS Press, Dec. 1, 2020. ISBN: 978-1-64368-150-4 978-1-64368-151-1. DOI: 10.3233/FAIA200844. URL: <http://ebooks.iospress.nl/doi/10.3233/FAIA200844> (visited on 11/23/2024), at 9; See also Sanjay Modgil and Henry Prakken. "The ASPIC+framework for Structured Argumentation: A Tutorial". In: *Argument & Computation* 5.1 (Jan. 2, 2014), pp. 31–62. ISSN: 1946-2166, 1946-2174. DOI: 10.1080/19462166.2013.869766. URL: <http://content.iospress.com/doi/10.1080/19462166.2013.869766> (visited on 11/27/2023).

²⁴Hanif Bhuiyan et al. "A Methodology for Encoding Regulatory Rules". In: *Proceedings of the 4th International Workshop on Mining and Reasoning with Legal Texts Co-Located with the 32nd International Conference on Legal Knowledge and Information Systems (JURIX 2019)*. International Workshop on Mining and Reasoning with Legal Texts 2019. Vol. 2632. Madrid, Spain: Rheinisch-Westfälische Technische Hochschule Aachen, Dec. 11, 2019, at 2.

rule, is represented as Y . The strength of the relationship between the premises and conclusion allows us to differentiate between strict rules, defeasible rules, and defeaters:²⁵

Strict rules (encoded as $X_1, \dots, X_n \rightarrow Y$) are inferences in the classical propositional sense. If the premise is indisputable, then so is the conclusion. E.g. "A Corporation is a Supplier":

$$\text{Corporation}(\text{Acme Inc.}) \rightarrow \text{Supplier}(\text{Acme Inc.}) \quad (12)$$

Defeasible rules (encoded as $X_1, \dots, X_n \Rightarrow Y$) are inferences that are generally true, but can be defeated by other information. An example in the guidelines is that a Supplier cannot be an exclusive provider unless the economic sector allows for a natural monopoly:

$$\text{Corporation}(\text{Acme Inc.}) \Rightarrow \text{ExclusiveSupplier}(\text{Acme Inc.}) \quad (13)$$

From this, we can conclude that a corporation can be an exclusive supplier, unless there is evidence to the contrary.

Defeaters (encoded as $X_1, \dots, X_n \rightsquigarrow Y$) are rules that can prevent a conclusion. Building on the previous example, we can maintain that:

²⁵Bhuiyan et al., "A Methodology for Encoding Regulatory Rules", at 8-9.

$$\neg(\text{Sector_AllowsNaturalMonopoly}(\text{Acme Inc.})) \rightsquigarrow \neg\text{ExclusiveSupplier}(\text{Acme Inc.}) \quad (14)$$

191

192 Defeasible logic can resolve conflicting information by allowing the prioritization
 193 of rules through the superiority ($\succ$) relation. E.g. $r1 \succ r0$ means that rule $r1$ takes
 194 precedence over rule $r0$. This can be used to resolve conflicts between rules, or to
 195 determine the applicability of a rule in a given context.

196 Finally, DDL takes into account deontic properties such as Obligation (O), Permis-
 197 sion (P) and Prohibition (F) and their relationships in SDL. For example, as to the
 198 attribute **ExclusiveSupplier**, the Prohibition against acting an exclusive supplier is
 199 equivalent to the Obligation not to act as an exclusive supplier.

$$[F]\text{ExclusiveSupplier} \equiv [O]\neg\text{ExclusiveSupplier} \quad (15)$$

200

201 Thus. the rule that disallows exclusive suppliers (subject to the exceptions for
 202 natural monopolies) can expressed as:

$$\emptyset(\text{Empty Set}) \Rightarrow [F] \text{ ExclusiveSupplier} \quad (16)$$

$$(\text{Sector_AllowsNaturalMonopoly}(\text{Acme Inc.})) \Rightarrow [P] \text{ Exclusive Supplier} \quad (17)$$

203

1.5 Next: Encoding Into LegalRuleML

All the logical systems discussed above build on each other and allow us to have a formalized representation of legal norms, enabling various operations and evaluations on these norms. Efficient, automated reasoning with these norms can be achieved by applying the logical model into a machine-readable format. The interest from the Artificial Intelligence and Law communities computational representation of norms has led to the development of digital formats for encoding the logical aspect of legal texts, such as the Rule Markup Language (RuleML),²⁶ Semantic Web Rule Language (SWRL), Rule Interchange Format (RIF), and the Legal Knowledge Interchange Format (LKIF).

LegalRuleML, an XML-based standard developed and maintained under the auspices of the Organization for the Advancement of Structured Information Standards (OASIS),²⁷ represents a convergence of many of these previous efforts, with broad support from both industry and academic communities.²⁸ LegalRuleML allows for the modelling of both constitutive rules and prescriptive rules as if-then statements (antecedent and consequent) with deontic effects, as well as properties and operations related to defeasibility. Detailed discussion of LegalRuleML's features (e.g. reification, temporal management, ontology references) will be provided as they are implemented in encoding the competition impact assessment guidelines. Besides the rich set of modern features, and a design approach that can accommodate multiple theo-

²⁶W3C. *RuleML - W3C RIF-WG Wiki*. 2005. URL: <https://www.w3.org/2005/rules/wg/wiki/RuleML> (visited on 01/12/2025).

²⁷Oasis Open. *LegalRuleML Core Specification Version 1.0*. Aug. 30, 2021. URL: <http://docs.oasis-open.org/legalruleml/legalruleml-core-spec/v1.0/legalruleml-core-spec-v1.0.html> (visited on 10/06/2023).

²⁸Tara Athan et al. "OASIS LegalRuleML". in: *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Law*. ICAIL '13: International Conference on Artificial Intelligence and Law. Rome Italy: ACM, June 10, 2013, pp. 3–12. ISBN: 978-1-4503-2080-1. DOI: 10.1145/2514601.2514603. URL: <https://dl.acm.org/doi/10.1145/2514601.2514603> (visited on 12/16/2024).

224 ries of logic and norms, there are practical advantages to employing LegalRuleML:1.
225 It is an open standard, with the full specification and documentation available online;
226 2. It has broader support compared to other formats, leading to a larger codebase of
227 examples and related applications; 3. As an XML-based format, it can be connected
228 to any ontology, readily providing the rules with semantics.

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